THE STRUCTURE OF MATHEMATICAL REALITY

Phase II — Chapter 6

The Unreasonable Effectiveness Problem

Assignment

Self-Study Curriculum

Write a two-page structured argument (with sections, not flowing prose) answering the question: "What is actually happening when a mathematical model makes an accurate prediction about a physical system?"

Your argument should:

1. State what a model is (structurally, not colloquially)
2. State what a prediction is (in terms of the model's operations)
3. Explain what it means for the prediction to be "accurate"
4. Propose a structural explanation for why accuracy is possible at all
5. Identify what remains unexplained by your proposal

   This is the central question of the curriculum. Your answer will be preliminary. That is fine: it should be revisited at the end.